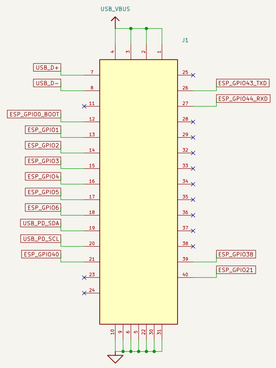
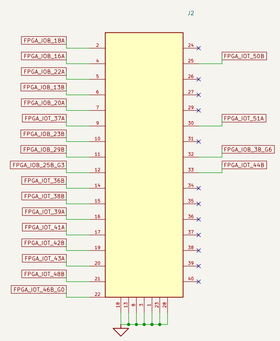
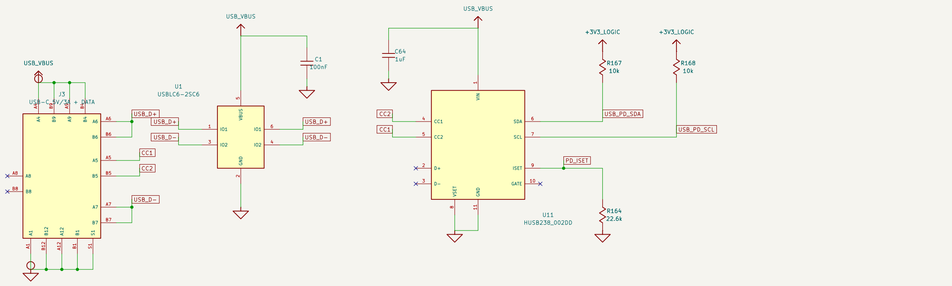
MODULE v1 / exact programmer mating interface

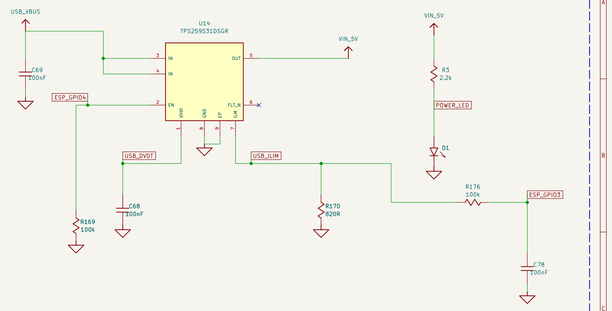




25 mm mating spacing / module top flush / no reset or boot switch

USB-C PD / 5 V, 3 A sink and protected power

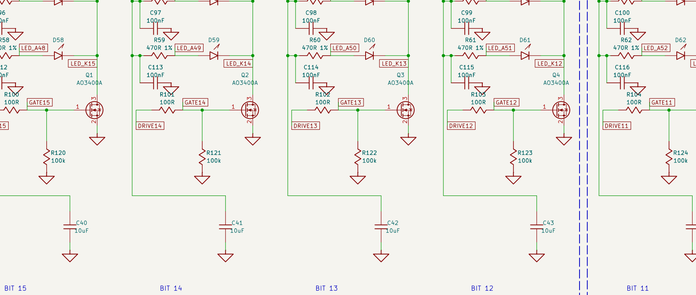




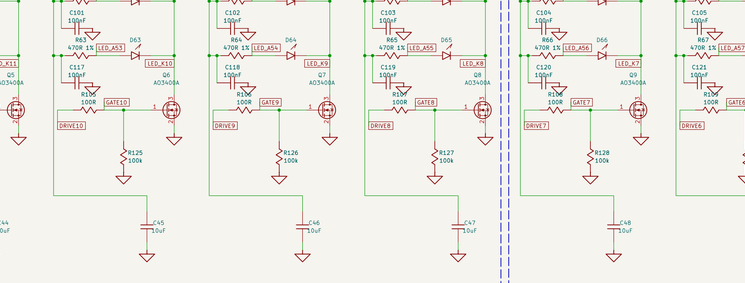
ESP_GPIO4 enables loads: R20R sets ~2.4A A; ADC: 0.22R V/A nominal, switched loads only.

Module and logic boot from raw 5 V. GPIO7/8 read PD status over I2C; firmware budgets current before enabling loads.

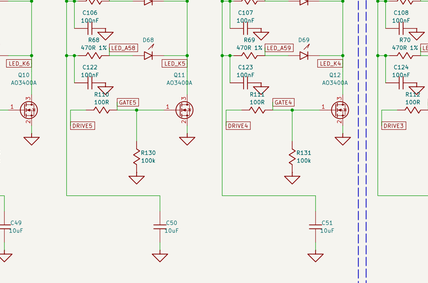
GNSS / PPS reference and active antenna bias



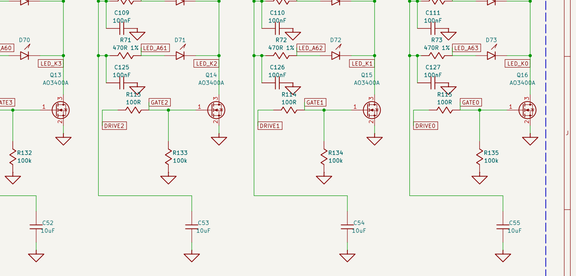
Trigger input / 5 V TTL / threshold ~2.5 V



Trigger output / 5 V TTL



10 MHz OCXO / 3.3 V linear supply

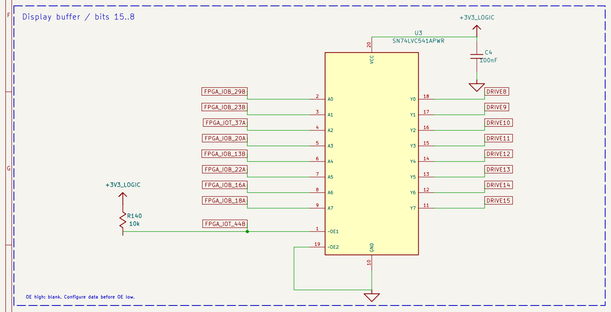


SOT-223 heat spreader below OCXO; GPIO5 enables.

2.1 W max warm-up; ~1.1 W LDO loss at 5 V: allow 3 min.

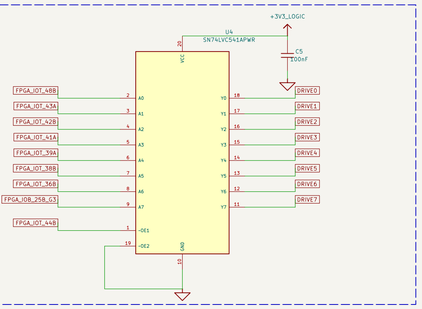
Manual-fit ASCO7FAJC-10.0000; Clock -> J2.32 / FPGA G6.

Display buffer / bits 15..8



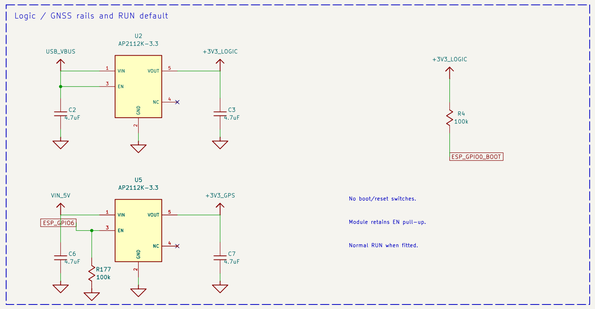
OE high: blank. Configure data before OE low.

Display buffer / bits 7..0

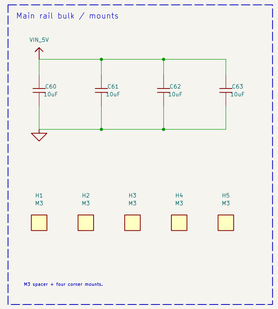


OE high: blank. Configure data before OE low.

Logic / GNSS rails and RUN default

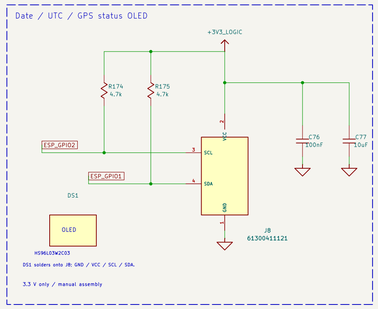


Main rail bulk / mounts



M3 spacer + four corner mounts.

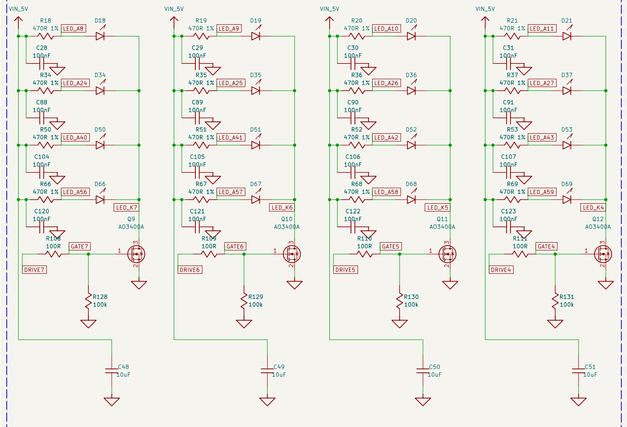
Date / UTC / GPS status OLED



OPTICAL BANK 1 / bits 15..12 / four identical rows

OPTICAL BANK 2 / bits 11..8 / four identical rows

OPTICAL BANK 3 / bits 7..4 / four identical rows



OPTICAL BANK 4 / bits 3..0 / four identical rows

Diagnostic probe pads / functional names on PCB / bare copper, no assembly parts

TP1 / VBUS — USB_VBUS
TP2 / 5V_LOAD — VIN_5V
TP3 / 5V_LOGIC — +5V_LOGIC
TP4 / 3V3_GPS — +3V3_GPS
TP5 / 3V3_OCXO — +3V3_OCXO
TP6 / 3V3_LOGIC — +3V3_LOGIC
TP7 / LOAD_EN — ESP_GPIO6
TP8 / PPS_LED — ESP_GPIO45
TP9 / GPS_EN — ESP_GPIO46
TP10 / PD_SDA — USB_PD_SDA
TP11 / PD_SCL — USB_PD_SCL
TP12 / LNDN — ESP_GPIO3
TP13 / PPS_SRC — GPS_PPS
TP14 / PPS_FPGA — FPGA_IOT_44B_G5
TP15 / PPS_ESP — ESP_GPIO38
TP16 / CLK_FPGA — FPGA_IOB_3B_G6
TP17 / TRIG_3V3 — TRIGGER_3V3
TP18 / TRIG_CMP — TRIGGER_CMP
TP19 / TRIG_5V3 — TRIGGER_5V3
TP20 / OUT_FPGA — FPGA_IOT_51A
TP21 / OUT_DRV — TRIGGER_DRV
TP22 / GPS_TX — ESP_GPIO43_TXD
TP23 / GPS_RX — ESP_GPIO44_RXD
TP24 / GPS_RST_N — GPS_RESET_N
TP25 / LED_OE_N — FPGA_IOT_41A
TP26 / GPS_BK — ESP_GPIO45_TXD
TP27 / GPS_RST_N — GPS_RESET_N
TP28 / OLED_SCL — ESP_GPIO2
TP29 / OLED_SDA — ESP_GPIO1
TP30 / BOOT_N — ESP_GPIO0_BOOT

bp-test
Sheet: /
File: calibrator.kicad_sch

Title: GPS optical time calibrator / module v1 carrier

Size: A1 | Date: | Rev: A draft
KiCad E.D.A. 10.0.6 | Id: 1/1